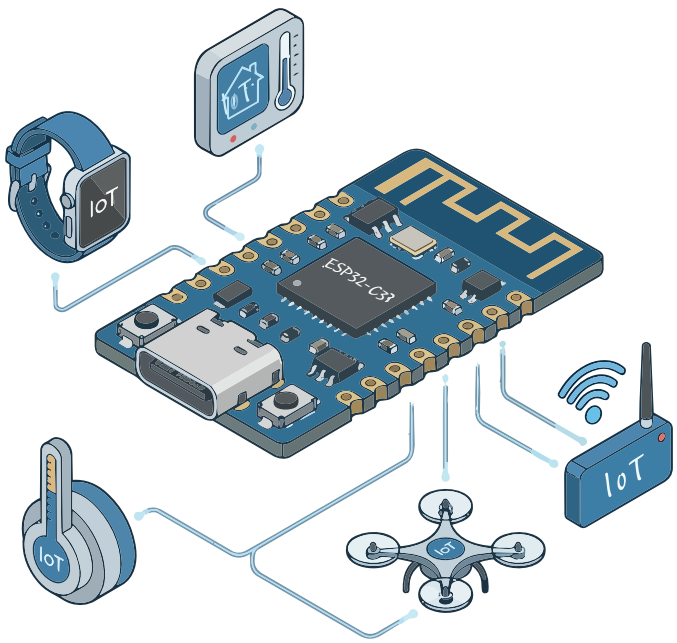
Creating Cutting-Edge IoT Devices with the ESP32 C3 Super Mini Board

The ESP32 C3 Super Mini board is a smaller but powerful alternative to the Raspberry Pi Pico boards. The MicroPython code presented in this hands-on guide on the ESP32 C3 board (and its interfacing with real world sensors) also works on the Pi Pico board. Everything has been tested on an Ubuntu LTS laptop with a Docker engine installed on it.



The ESP32 C3 Super Mini board is a highly economical thumb-sized (less than half in size when compared with any Pi Pico board) IoT powerhouse. It is driven by a 32-bit RISC-V single-core processor, running up to 160MHz with 4MB onboard Flash memory and 400KB SRAM. It comes with 2.4GHz Wi-Fi (802.11 b/g/n, up to 150 Mbps) and Bluetooth 5.0 LE (including Bluetooth mesh) capabilities. The development board is powered by 5V via USB Type-C or 3.3V–6V external input. The price of the Super Mini board with these impressive specs is half that of a Pi Pico board. You could purchase it online through various sites for around Rs 250 to Rs 270 with or without headers soldered.

Figure 1 shows what the board looks like. The power on red LED (left up), onboard blue LED (right down) and reset button (right up) are clearly marked. It is a 16-pin board (8 on each side) with 11 accessible GPIOs providing two 12-bit SAR ADCs up to six channels, six PWM channels, three SPI interfaces, an I2C interface, an I2S interface, two UART interfaces, etc. You can drive both 5V as well as 3.3V devices like sensors through the board. The pin diagram of the board is shown in Figure 2, which you should keep handy to follow for wiring various sensors and the MicroPython code shown in this article.

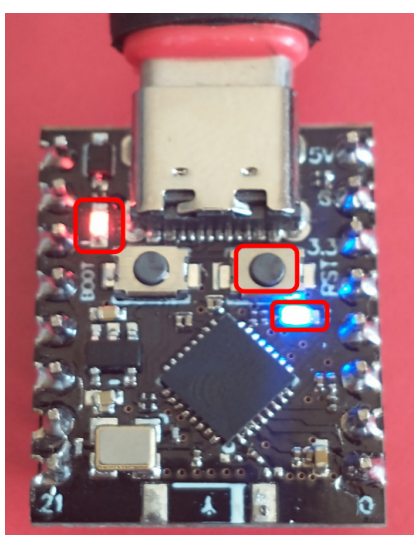


Figure 1: How the ESP32 C3 Super Mini looks

The board is fully covered by the MicroPython firmware and is totally driven by Thonny IDE and the command line pyboard and mpremote tools introduced in my past articles in *OSFY*.

Ok, time to lay our hands on the Super Mini board to see it in real world physical action. First, we need to flash the board with an appropriate version of the MicroPython firmware. This